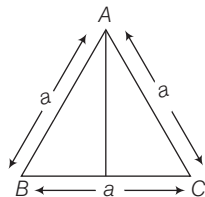


(v) The radius of circumcircle = $\frac{a}{\sqrt{3}}$



EXAMPLE 15. The perimeter of an equilateral triangle whose area is $4\sqrt{3}$ cm² is

- a. 4 cm b. 3 cm c. 12 cm d. 7 cm

Sol. c. Area of an equilateral triangle = $\frac{\sqrt{3}}{4} (\text{Side})^2$

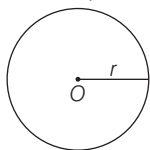
$$\Rightarrow \frac{\sqrt{3}}{4} (\text{Side})^2 = 4\sqrt{3}, (\text{Side})^2 = 16$$

$$\therefore \text{Side} = 4 \text{ cm}$$

$$\text{Hence, the perimeter of an equilateral triangle} \\ = 3 \times \text{Side} = 3 \times 4 = 12 \text{ cm}$$

CIRCLE

Let the radius of a circle be r , then



- (i) Circumference of circle = $2\pi r = \pi D$ units
[$\because D$ is diameter, $D = 2r$]
(ii) Area of circle = πr^2 sq units
(iii) Distance covered by a wheel in one revolution
= Circumference of the wheel

EXAMPLE 16. The circumference of a circle whose area is 24.64 m² is

- a. 17.2 m b. 17.4 m c. 17.6 m d. 18.0 m

Sol. c. Let the radius of the circle be r m.

$$\therefore \pi r^2 = 24.64$$

$$\Rightarrow \frac{22}{7} r^2 = 24.64 \Rightarrow r^2 = \frac{7 \times 24.64}{22}$$

$$\Rightarrow r = \sqrt{\frac{7 \times 24.64}{22}} = 2.8$$

$$\text{Thus, circumference} = 2\pi r = 2 \times \frac{22}{7} \times 2.8 = 17.6 \text{ m}$$

EXAMPLE 17. The diameter of the driving wheel of a car is 140 cm. Then, in order to keep the speed of 66 km/h, how many revolutions per minute must the wheel make?

- a. 250 b. 275 c. 290 d. 295

Sol. a. Given, radius of the wheel = 70 cm = 0.7 m

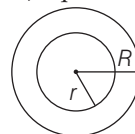
$$\text{Circumference of the wheel} = 2\pi r = \left(2 \times \frac{22}{7} \times 0.7\right) = 4.4 \text{ m}$$

$$\text{Distance to be covered in 1 min} = \frac{66 \times 1000}{60} = 1100 \text{ m}$$

$$\therefore \text{Number of revolutions per minute} = \frac{1100}{4.4} = 250$$

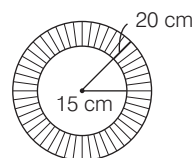
Circular Ring

If ' R ' and ' r ' be outer and inner radii of a ring, then the area of ring = $\pi(R^2 - r^2)$ sq units



EXAMPLE 18. The area of a ring whose outer and inner radii are respectively 20 cm and 15 cm is

- a. 440 cm² b. 550 cm² c. 565 cm² d. 675 cm²



Sol. b. Given, radius of outer circle (R) = 20 cm

Radius of inner circle (r) = 15 cm

$$\therefore \text{Area of the ring} = [\text{Area of outer circle} \\ - \text{Area of inner circle}]$$

$$= (\pi R^2 - \pi r^2) = \pi(R^2 - r^2) = \frac{22}{7}(20^2 - 15^2)$$

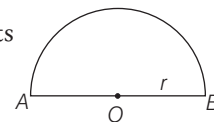
$$= \frac{22}{7}(20 + 15)(20 - 15) = \frac{22}{7} \times 35 \times 5 = 550 \text{ cm}^2$$

Semi-circle

A diameter divides a circle into two equal parts. Each of these two arcs is called semi-circle. If r is the radius of a circle, then

(i) Area of semi-circle = $\frac{1}{2} \pi r^2$ sq units

(ii) Perimeter of semi-circle
= $(\pi r + 2r) = (\pi r + D)$ units



EXAMPLE 19. If the perimeter of a semi-circular protractor is 36 cm, then its diameter is

- a. 6 cm b. 7 cm c. 7.5 cm d. 14 cm

Sol. d. Let the radius of the protractor be r cm, then perimeter

$$= (\pi r + 2r) = (\pi + 2)r = \frac{36}{7}r \Rightarrow 36 = \frac{36r}{7}$$

$$\therefore r = 7 \text{ cm}$$

$$\text{Hence, diameter of the protractor} = 2r = 2 \times 7 = 14 \text{ cm}$$

Quadrant of a Circle

If r is the radius of a circle, then

$$\begin{aligned} \text{(i) The perimeter of the quadrant} \\ &= \frac{1}{4} (\text{Circumference of a circle}) + 2r \\ &= \frac{1}{4} \times 2\pi r + 2r = \left(\frac{\pi}{2} + 2\right)r \text{ units} \end{aligned}$$

$$\text{(ii) Area of the quadrant} = \frac{1}{4} (\text{Area of circle}) = \frac{\pi r^2}{4} \text{ sq units}$$

If two diameters are perpendicular to each other, then they divide the circle into four quadrants.

Area of Sector

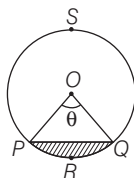
If θ be the angle at the centre of a circle of radius r , then

$$\text{(i) Length of the arc } \overline{PQ} = \frac{2\pi r \theta}{360^\circ} = \frac{\pi r \theta}{180^\circ}$$

$$\text{(ii) Area of sector } OPRQO = \frac{\pi r^2 \theta}{360^\circ}$$

$$\begin{aligned} \text{(iii) Area of segment } PRQP \\ &= (\text{Area of sector } OPRQO) - \text{Area of } \triangle OPQ \\ &= \frac{\pi r^2 \theta}{360^\circ} - \frac{1}{2} r^2 \sin \theta \end{aligned}$$

$$\text{(iv) Area of major segment } QPSQ = (\text{Area of circle}) - (\text{Area of minor segment } PRQP)$$



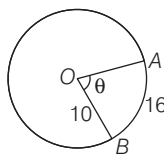
EXAMPLE 20. The arc AB of the circle with centre at O and radius 10 cm has length 16 cm. What is the area of the sector bounded by the radii OA , OB and the arc AB ?

- a. $40\pi \text{ cm}^2$ b. 40 cm^2 c. 80 cm^2 d. $20\pi \text{ cm}^2$

Sol. c. Arc length $= 2\pi r \cdot \frac{\theta}{360^\circ}$

$$\Rightarrow 16 = 2\pi r \cdot \frac{\theta}{360^\circ} \Rightarrow \frac{\theta}{360^\circ} = \frac{16}{2\pi r}$$

$$\begin{aligned} \text{Now, area of sector } OABO &= \pi r^2 \cdot \frac{\theta}{360^\circ} \\ &= \pi r^2 \cdot \frac{16}{2\pi r} = 8r = 8 \times 10 = 80 \text{ cm}^2 \end{aligned}$$



REGULAR POLYGON

A polygon that has all sides equal and all interior angles equal are regular polygon.

Let, n = number of sides, a = length of side
 r = radius of inscribed circle (or inradius)
 R = radius of circumcircle

$$\begin{aligned} \text{Area of polygon} &= \frac{1}{4} na^2 \cot\left(\frac{180^\circ}{n}\right) \\ &= nr^2 \tan\left(\frac{180^\circ}{n}\right) = \frac{1}{2} nR^2 \sin\left(\frac{360^\circ}{n}\right) \end{aligned}$$

$$\text{Perimeter of polygon} = n \times a = 2nr \sin\left(\frac{\pi}{n}\right)$$

$$\text{Inradius of polygon, } r = \frac{1}{2} a \cot\left(\frac{180^\circ}{n}\right)$$

$$\text{circumradius of polygon, } R = \frac{1}{2} a \operatorname{cosec}\left(\frac{180^\circ}{n}\right)$$

$$\text{(i) Area of regular pentagon} = \frac{1}{4} \sqrt{5(5+2\sqrt{5})} a^2 \text{ sq units}$$

$$\text{(ii) Area of regular hexagon} = \frac{3\sqrt{3}a^2}{2} \text{ sq units}$$

$$\text{(iii) Area of regular octagon} = 2(\sqrt{2}+1)a^2 \text{ sq units}$$

EXAMPLE 21. If the area of a regular hexagon is $96\sqrt{3} \text{ cm}^2$, then its perimeter is

- a. 36 cm b. 48 cm c. 54 cm d. 64 cm

Sol. b. Given that, area of a regular hexagon $= 96\sqrt{3}$
 $\Rightarrow \frac{3}{2} \sqrt{3} (\text{Side})^2 = 96\sqrt{3} \Rightarrow (\text{Side})^2 = 64 = (8)^2$

$$\therefore \text{Side} = 8 \text{ cm}$$

$$\begin{aligned} \text{Hence, perimeter of a regular hexagon} \\ &= 6 \times (\text{Side}) = 6 \times 8 = 48 \text{ cm} \end{aligned}$$

Some Useful Results

- Angle inscribed by minute hand in 60 min $= 360^\circ$
- Angle inscribed by hour hand in 12 h $= 360^\circ$
- Angle inscribed by minute hand in 1 min $= 6^\circ$
- If the length of a square/rectangle is increased/decreased by $x\%$ and the breadth is increased/decreased by $y\%$, the net effect on the area is given by

$$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right] \%$$

- If the length of a square/rectangle is increased/decreased by $x\%$ and the breadth is increased/decreased by $y\%$ the net effect on the area is given by

$$\text{Net effect} = \left[\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right] \%$$

Note (+) Increase
 (–) Decrease

- If the side of a square/rectangle/triangle is doubled the area is increased by 300%, i.e. the area becomes four times of itself.
- If the radius of a circle is decreased by $x\%$, then, area is decreased by $\left(-2x + \frac{x^2}{100}\right)\%$. And, if radius is increased

$$\text{by } x\%, \text{ then area is increased by } \left(2x + \frac{x^2}{100}\right)\%.$$

- If a room of dimensions $(l \times b)$ m is to be paved with square tiles, then

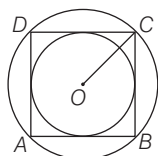
(i) the side of the largest square tile = HCF of l and b

(ii) the least number of tiles required

$$= \frac{l \times b}{(\text{HCF of } l \text{ and } b)^2}$$

- Area of a square inscribed in a circle of radius r is $2r^2$ and the side of a square inscribed in a circle of radius r is $\sqrt{2}r$.
- The area of the largest triangle inscribed in a semi-circle of radius r is r^2 .

EXAMPLE 22. The ratio of the areas of the incircle and circumcircle of a square are



- a. 1 : 1 b. 2 : 1 c. 1 : 2 d. 3 : 1

Sol. c. Let side $AB = BC = CD = AD = x$

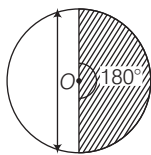
$$\therefore \text{Diagonal of square} = \sqrt{2}x$$

$$\therefore \text{Radius of circumcircle} = \frac{\sqrt{2}x}{2} = \frac{x}{\sqrt{2}}$$

$$\therefore \text{Radius of incircle} = \frac{x}{2}$$

$$\therefore \text{Required ratio} = \left(\frac{\pi x^2}{4} : \frac{\pi x^2}{2} \right) = 2 : 4 = 1 : 2$$

EXAMPLE 23. The minute hand of a clock is 14 cm long. The area of the face of the clock inscribed by the minute hand in 30 min is



- a. 308 cm^2 b. 312 cm^2 c. 412 cm^2 d. 416 cm^2

Sol. a. Angle inscribed by the minute hand in 60 min = 360°

Angle inscribed by the minute hand in 30 min

$$= \frac{360^\circ}{60} \times 30 = 180^\circ$$

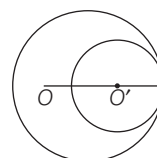
So, $\theta = 180^\circ$ and $r = 14$ cm.

Required area swept by minute hand in 30 min

$$= \frac{\theta}{360^\circ} \times \pi r^2 = \frac{180^\circ}{360^\circ} \times \frac{22}{7} \times 14 \times 14$$

$$= \frac{1}{2} \times \frac{22}{7} \times 14 \times 14 = 308 \text{ cm}^2$$

EXAMPLE 24. Two circles touch internally. The sum of their areas is $116\pi \text{ cm}^2$ and distance between their centres is 6 cm. Then, the radii of the circles are



- a. 4 cm and 9 cm
b. 5 cm and 10 cm
c. 4 cm and 8 cm
d. 4 cm and 10 cm

Sol. d. Let the radius of outer circle be R and radius of inner circle = r cm

$$\therefore \text{Given, } \pi R^2 + \pi r^2 = 116\pi, \quad R^2 + r^2 = 116$$

If O and O' be the centre of these circles, then

$$OO' = (R - r)$$

Also, $(R - r) = 6$ [given]

$$\text{So, } (R + r)^2 + (R - r)^2 = 2(R^2 + r^2)$$

$$\Rightarrow (R + r)^2 = 2(116) - 36 = 196$$

$$\Rightarrow R + r = \sqrt{196}$$

$$\text{So } R + r = 14 \quad \dots(i)$$

$$R - r = 6 \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$r = 4 \text{ cm and } R = 10 \text{ cm}$$

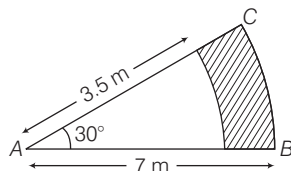
> PRACTICE EXERCISE

1. The diagonal of a square field measures 50 m. The area of square field is
(a) 1250 m^2 (b) 1200 m^2 (c) 1205 m^2 (d) 1025 m^2
2. The length of a rectangle is 2 cm more than its breadth. The perimeter is 48 cm. The area of the rectangle (in cm^2) is
(a) 96 (b) 128 (c) 143 (d) 144
3. In a circle of radius 42 cm, an arc subtends an angle of 72° at the centre. The length of the arc is
(a) 52.8 cm (b) 53.8 cm (c) 72.8 cm (d) 79.8 cm
4. An isosceles right angle triangle has area 200 cm^2 . The length of its hypotenuse is
(a) $15\sqrt{2}\text{ cm}$ (b) $\frac{10}{\sqrt{2}}\text{ cm}$ (c) $10\sqrt{2}\text{ cm}$ (d) $20\sqrt{2}\text{ cm}$
5. With in a rectangular garden 10 m wide and 20 m long, we wish to pave a walk around the borders of uniform width so as to leave an area of 96 m^2 for flowers. The width of the walk is
(a) 1 m (b) 2 m (c) 2.5 m (d) 2.56 m
6. The least number of square slabs that can be fitted in a room 10.5 m long and 3 m wide, is
(a) 12 (b) 13 (c) 14 (d) 15
7. If the side of a square be increased by 50%, the percent increase in area is
(a) 50 (b) 100 (c) 125 (d) 150
8. The area of the largest circle that can be drawn inside a square of side 14 cm in length, is
(a) 84 cm^2 (b) 96 cm^2 (c) 104 cm^2 (d) 154 cm^2
9. If the radius of a circle is decreased by 50%, its area will decrease by
(a) 25% (b) 50% (c) 75% (d) 100%
10. The area of the circle whose circumference is equal to the perimeter of a square of side 11 cm is
(a) 154 cm^2 (b) 144 cm^2 (c) 134 cm^2 (d) 124 cm^2
11. A wire is in the form of a circle of radius 42 cm. It is bent into a square. The side of the square is
(a) 33 cm (b) 66 cm (c) 78 cm (d) 112 cm
12. A horse is tied to a pole with 28 m long string. The area which the horse can graze is equal to
(a) 246 m^2 (b) 2404 m^2 (c) 2464 m^2 (d) 2164 m^2
13. The area of ring is 418 cm^2 . If the radius of the smaller circle is 6 cm. The radius of the bigger circle is
(a) 18 cm (b) 16 cm (c) 13 cm (d) 10 cm
14. The length of a rectangle is increased by 60%. By what per cent would the width have to be decreased to maintain the same area?
(a) $37\frac{1}{2}\%$ (b) 60% (c) 75% (d) 120%
15. The perimeter of a rectangular field is 240 m and the ratio between the length and breadth is 5 : 3. The area of the field is
(a) 33750 m^2 (b) 3375 m^2 (c) 3500 m^2 (d) 3950 m^2
16. The inner circumference of a circular park is 440 m. The track is 14 m wide. The diameter of the outer circle of the track is
(a) 168 m (b) 169 m (c) 144 m (d) 108 m
17. If the length and breadth of a rectangular plot are increased by 50% and 20% respectively, then the new area is how many times the original area?
(a) $\frac{5}{9}$ (b) 10 (c) $\frac{9}{5}$ (d) None of these
18. The area of an isosceles triangle, each of whose equal sides is 13 cm and whose base is 24 cm is
(a) 60 cm^2 (b) 55 cm^2 (c) 50 cm^2 (d) 40 cm^2
19. The difference between the sides at right angles in a right angled triangle is 14 cm. The area of the triangle is 120 cm^2 . The perimeter of the triangle
(a) 68 cm (b) 64 cm (c) 60 cm (d) 58 cm
20. In a four sided field, the length of the longer diagonal is 128 m. The lengths of the perpendicular from the opposite vertices upon this diagonal are 22.7 m and 17.3 m. The area of the field is
(a) 2246 m^2 (b) 2460 m^2 (c) 2540 m^2 (d) 2560 m^2
21. The adjacent sides of a parallelogram are 36 cm and 27 cm in length. If the distance between the shorter sides is 12 cm. The distance between the longer sides is
(a) 9 cm (b) 10 cm (c) 11 cm (d) 12 cm
22. The area of the quadrilateral whose sides measures are 9 cm, 40 cm, 28 cm and 15 cm and in which the angle between the first two sides is a right angle is
(a) 206 cm^2 (b) 306 cm^2 (c) 356 cm^2 (d) 380 cm^2
23. A bicycle wheel makes 5000 revolutions in moving 11 km. The diameter of the wheel is
(a) 50 cm (b) 60 cm (c) 70 cm (d) 80 cm

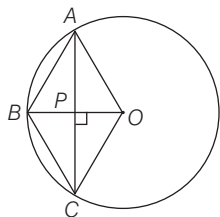
- 24.** Two circles touch externally. The sum of their areas is $130\pi\text{ cm}^2$ and the distance between their centres is 14 cm. The radii of the circles are
 (a) 11 cm, 3 cm (b) 10 cm, 4 cm
 (c) 9 cm, 5 cm (d) 8 cm, 6 cm

- 25.** The minute hand of a clock is 12 cm long. The area of the face of the clock described by the minute hand in 35 min.
 (a) 284 cm^2 (b) 294 cm^2 (c) 274 cm^2 (d) 264 cm^2

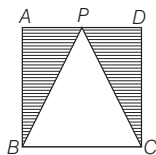
- 26.** In the given figure, sectors of two concentric circles of radii 7 cm and 3.5 cm are shown. The area of the shaded region



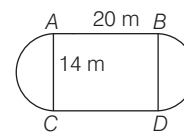
- (a) $\frac{77}{4}\text{ cm}^2$ (b) $\frac{77}{8}\text{ cm}^2$
 (c) $\frac{77}{2}\text{ cm}^2$ (d) None of these
- 27.** Of the two square fields, the area of one is 1 hectare, while the other one is broader by 2%. The difference in their areas is
 (a) 104 m^2 (b) 200 m^2 (c) 204 m^2 (d) 404 m^2
- 28.** If the diameter of the circle is increased by 100%, its area is increased by
 (a) 100% (b) 200% (c) 300% (d) 400%
- 29.** If the two parallel sides of a trapezium are 15 cm and 25 cm respectively and the distance between them is 7 cm, then the area of the trapezium is
 (a) 105 cm^2 (b) 125 cm^2
 (c) 140 cm^2 (d) None of these
- 30.** The diagonals of a rhombus are 24 cm and 10 cm, respectively. The perimeter of the rhombus is
 (a) 50 cm (b) 52 cm (c) 60 cm (d) 68 cm
- 31.** In the given figure $OABC$ is a rhombus whose three vertices A, B, C lie on the circle of radius 10 cm. The area of rhombus is



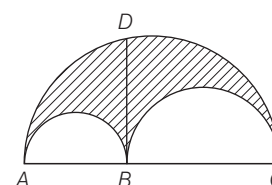
- (a) $50\sqrt{3}\text{ cm}^2$ (b) $100\sqrt{3}\text{ cm}^2$ (c) $75\sqrt{3}\text{ cm}^2$ (d) $125\sqrt{3}\text{ cm}^2$
- 32.** In the adjoining figure, $ABCD$ is a rectangle such that $AB = 2AD = a$. P is the mid-point of AD . The area of the shaded region is
 (a) $\frac{1}{3}a^2$ (b) $\frac{1}{2}a^2$ (c) $\frac{1}{4}a^2$ (d) a^2



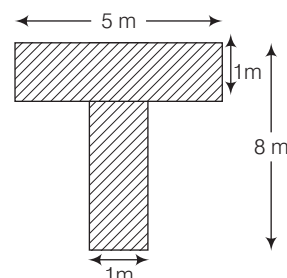
- 33.** A garden is in the form of a rectangle with semi-circular ends on the either side as shown in the diagram below. The length and breadth of the rectangle are 20 m and 14 m, respectively. The cost of levelling the plot at ₹ 25
 (a) ₹ 10850 (b) ₹ 5425 (c) ₹ 8510 (d) ₹ 4255



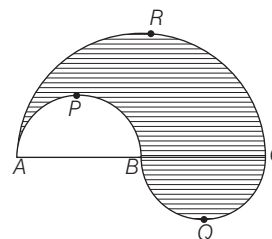
- 34.** Consider the adjoining figure, let $AB = 4\text{ cm}$, $BC = 14\text{ cm}$, then the area (shaded) bounded by three semi-circles as shown in the adjoining figure in cm^2 , is π times



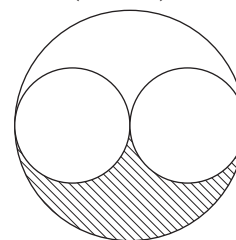
- (a) 48 (b) 24 (c) 14 (d) 12
- 35.** Area of shaded portion as shown in the figure is



- (a) 10 m^2 (b) 12 m^2 (c) 14 m^2 (d) 16 m^2
- 36.** The boundary of the shaded region in the adjoining diagram consists of three semi-circular arc, the smaller two being equal. If the diameter of the large one is 10 cm, then the length of the boundary is
 (a) 31 cm (b) $10\pi\text{ cm}$ (c) $20\pi\text{ cm}$ (d) $19\pi\text{ cm}$



- 37.** In the adjoining figure, the larger circle with radius 4 cm is touched internally by two smaller circles which also touch each other externally at the centre O of the larger circle. The area of shaded region is (in cm^2)



- (a) π (b) 2π (c) 3π (d) 4π